

Feature Understanding with Pre-trained CNN Models



What is this project about?

This project compares how well three CNN architectures — ResNet-101, GoogleNet, and ZFNet — represent images as embeddings, rather than training a new classifier. A strong CNN should produce embeddings where images of the same category sit close together, and different categories sit far apart.

Context

- Dataset: 1,000 images, 5 scene classes (Intel Image Classification)
- Classes: buildings, forest, glacier, mountain, sea
- Pre-trained CNNs used as general-purpose feature extractors
- Applications: image retrieval, clustering, similarity search, transfer learning

Objectives

What we set out to achieve — and did

01 Build a complete pipeline

Extract, compare, evaluate, and visualize embeddings from 3 CNN architectures end to end.

02 Measure nearest-neighbor consistency

Retrieve top-10 similar images per class and check how many share the query's true class.

03 Compare intra/inter-class distance

Quantify how tightly same-class images cluster vs. how far different classes separate.

04 Visualize with t-SNE

Project high-dimensional embeddings to 2D to visually confirm cluster structure.

05 Add linear probing accuracy

Train a simple classifier on frozen embeddings for a hard, quantitative comparison score.

Outcome: ResNet-101 and GoogleNet both produced strong, usable embeddings; ZFNet (random weights, no pretraining) was consistently weakest across every metric.

Related Work

The architectures and methods this project builds on



AlexNet / ZFNet

Krizhevsky et al. 2012 · Zeiler & Fergus 2013

Early deep CNNs for classification. ZFNet refined AlexNet and introduced feature visualization. Implemented from scratch here — no pretrained weights available.



GoogleNet (Inception)

Szegedy et al. 2014

Uses inception modules to capture multi-scale patterns efficiently. Used pretrained, with classifier head replaced by identity to output embeddings.



ResNet

He et al. 2015

Residual connections enable very deep, trainable networks. ResNet-101 used pretrained, producing 2048-D embeddings.



t-SNE

van der Maaten & Hinton 2008

Projects high-dimensional data to 2D while preserving local neighborhoods — used to visualize embedding cluster structure.

How this project differs: no new model or training method is proposed — known architectures are applied as fixed feature extractors and compared experimentally on identical data using four independent evaluation methods.

Methods — Data & Models

How the dataset and the three CNNs were used



Dataset & Preprocessing

Source: Intel Image Classification dataset

Classes: buildings, forest, glacier, mountain, sea

Size: 200 images/class → 1,000 images total, balanced

Resize: 224 × 224 pixels

Normalize: ImageNet mean/std (required for pretrained ResNet-101 & GoogleNet)



Models Used



ResNet-101

Pretrained · fc → Identity() · 2048-D embedding



GoogleNet

Pretrained · fc → Identity() · 1024-D embedding



ZFNet

Implemented from scratch · random weights · 4096-D embedding

Methods — The Pipeline

From raw images to final evaluation metrics



Batching: All 1,000 images processed per model in batches of 32, using `torch.no_grad()` during inference.

Saved artifacts: embeddings/ (vectors, labels, paths) → results/ (similarity + metrics + linear probe scores) → plots/ (t-SNE figures)

Similarity search: For each class, 1 query image → cosine similarity against all embeddings → top-10 nearest neighbors retrieved, per model.

Results — Metrics Comparison

Four independent evaluation methods, side by side

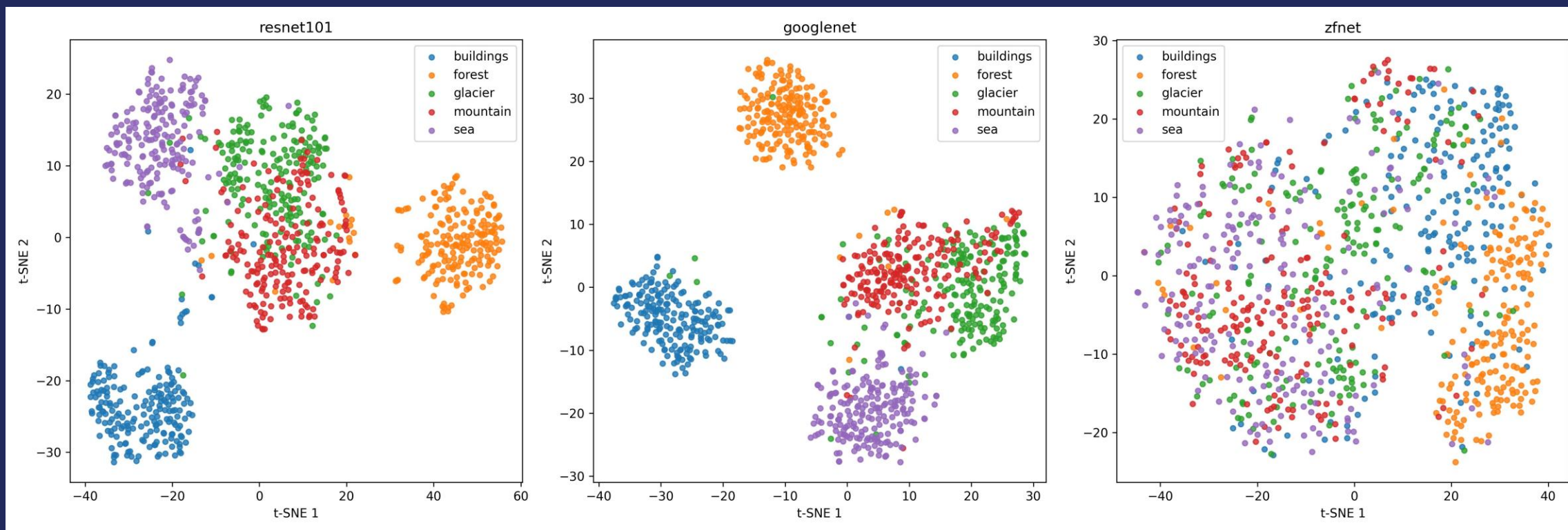
Model	NN Consistency	Intra-class Dist.	Inter-class Dist.	Separation Gap	Linear Probe Acc.
ResNet-101	0.70	0.5518	0.8031	0.2512	89.50%
GoogleNet	0.60	0.3330	0.4946	0.1616	90.50%
ZFNet	0.38	0.0274	0.0342	0.0068	59.50%

Key takeaways

- ResNet-101 leads on 3 of 4 metrics — nearest-neighbor consistency, separation gap, and t-SNE structure.
- GoogleNet edges ahead only on linear probe accuracy (90.5% vs 89.5%) — a near-tie, and a genuinely interesting result rather than a contradiction to hide.
- ZFNet is unambiguously last on every single metric — expected, since it has no pretrained weights.

Results — Visual Evidence

t-SNE projections confirm the numbers



ResNet-101: tight, well-separated clusters — especially forest, buildings, sea. **GoogLeNet:** visible groupings with more overlap. **ZFNet:** no meaningful structure — colors are mixed throughout, confirming its weak metric scores.